1. Knowledge-base construction

Validated corpus

Human-written
contract / test pairs

Code-aware chunking

Function and test units
Provenance + heuristic labels

Dual indexes

Dense + BM25 indexes
One entry per unit

Structural catalogue

Full text + extracted setup
Static test-to-function links

2. Retrieval-augmented test generation

Target contract

Source code
+ testing request

Query decomposition

One code query per
public / external function

Hybrid retrieval

Dense + BM25
RRF rank fusion

Cross-encoder reranking

Target function vs. candidate
Second RRF fusion

Function-to-test pivot

Retrieved functions → tests
Associated extracted setup

Context selection

Round-robin + deduplication
Top-K / character budget

Prompt assembly + LLM

Target + instructions + fixtures
Retrieved tests → Codestral

Generated unit tests

JavaScript test suite
Hardhat / Mocha / Chai

3. Execution and evaluation

Validation

JavaScript syntax check
Solidity compilation

Hardhat execution

Mocha + solidity-coverage
Target-contract coverage

Evaluation metrics

Fully passing suites; pass rate
Coverage + failure categories

Cost and traceability

Prompt size + retrieval time
Prompts, logs and reports

Input data

Processing module

Generated output

Stored data